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"So you got the brain but have you got the touch?": the impact of the social presence of service robot and service innovativeness on customer satisfaction

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ABSTRACT

While robots offer clear benefits in fulfilling functional tasks, service providers remain cautious when using robots in roles that include interaction with customers, as their social benefits may be questionable. This paper examines how the social presence of the robot makes service encounters with service robots satisfactory to customers. Across two online studies and one laboratory study, results show that social presence positively influences customer satisfaction with service encounters. Service innovativeness mediates this relationship, but only for more innovative customers. The results highlight the importance of balancing social presence and anthropomorphism in service robots, as robot anthropomorphism negatively moderates the relationship between social presence and satisfaction. The paper contributes to service robot research by explaining how social presence shapes customer responses and when its benefits vary. Managerially, the findings suggest that providers should implement robots high in social presence while carefully balancing their level of anthropomorphism to maximize their benefits.

1. Introduction

Robots can cook, clean, check inventory, serve food, do product demonstrations, or even serve as witnesses in criminal cases, as in the case of googly-eyed robot Marty (Jiang 2020). Service robots are not only versatile, but increasingly ubiquitous, as both big retail chains and smaller service providers are welcoming service robots. Walmart uses Bossa Nova robots for inventory checks, while in Moscow, Lenta supermarkets employ the Promobot robot to engage with customers and promote products. The Gran Caffé Rapallo in Italy and Naulo restaurant in Nepal employ robots to help serve orders to customers (Jiang 2020). Given the growing prevalence of service encounters with service robots, there is a need to better understand their interactions with customers. In this paper, I examine how the social presence of service robots influences customer satisfaction. Further, I investigate the mediating role of service innovativeness. Finally, to provide further depth to the understanding of the dynamics of those relationships, I consider the moderating role of two additional factors associated with the customer–robot encounter: robot anthropomorphism and customer innovativeness. Service robots are defined as "information technology in a physical embodiment, providing customized services by performing physical as well as nonphysical tasks with a high degree of autonomy"(Jörling et al. 2019 p.405). Service robots often need to go beyond performing mechanical tasks as they interact and connect with the customer (Blut et al. 2021). That is no easy task as in service encounters, service employees develop positive relationships with customers, building rapport, engagement, and trust, thereby delivering both emotional and social value (Wirtz et al. 2018). The social aspect of the service encounter plays a central role in enhancing customer experience (Brakus et al., 2009) and driving customer loyalty (Price & Arnould, 1999). Although scholars predicted that by 2025 service robots would engage customers on a social level (van Doorn et al. 2017) and emphasized the importance of feeling AI (Huang and Rust 2021), customers often remain wary and averse to them (Borghi et al. 2023; Čaić et al. 2019; Mende et al. 2019). While productivity gain of service robots is evident, there remains a need to understand how customers perceive service robots and evaluate services delivered by them (Grazzini et al. 2023; Yam et al. 2021).

Considering that even the mere presence of the human employee positively influences customer satisfaction (Söderlund 2016), it is challenging for service firms to benefit from robot–customer interaction on a similar level. Thus, robots mimic humans both in behavior and shape, as features and behavior resembling humans make robots easier to accept for customers (Damiano and Dumouchel 2018; Söderlund

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2023). A key aspect in the context of service robots is social presence (Čaić et al. 2020; van Doorn et al., 2017). Social presence refers to the extent to which an interaction partner, in this case the service robot, is experienced as a social actor that is, as an entity capable of participating in the interaction and responding in socially meaningful ways (van Doorn et al. 2017; Lee, 2004). Although social presence is inherent in face-to-face interactions between people, it is not necessarily the case for human-to-robot interactions (Biocca 1997; Kim et al. 2013). Nevertheless, customers perceive robots as social entities despite recognizing their nonhuman nature (van Doorn et al. 2017). Hence, the social presence of a robot may also play a role in customer satisfaction with the service encounter.

Service robots are not only functional improvements but also help consumers form beliefs and attitudes towards delivered services. In fact, service robots are often assumed to offer innovative services (Amelia et al. 2022; Chu et al. 2017), as they introduce some degree of newness or novelty for customers (Witell et al. 2016). Indeed, it is paramount that customers perceive services as innovative, as service innovativeness provides the customer with unique benefits and elevates differentiation, which, in turn, enhances the performance of the service provider (Liu 2013). I define service innovativeness as customer's evaluation of the meaningfulness and novelty of service relative to alternatives (Mu et al. 2018; Zolfagharian and Paswan 2008). Given that service evaluations are known to be antecedents of satisfaction (Taylor and Baker, 1994), service innovativeness may add to understanding how service robots influence consumer responses. However, prior research (except McLeay et al. 2021) has rarely considered how customers evaluate innovativeness of services provided by service robots or what characteristics of service robots influence service innovativeness.

This paper responds to the call for more research on robots' influence on customers (Lu et al. 2020) and technology-enabled service provision (Hollebeek et al. 2019). Although the literature provides a strong conceptual foundation for understanding service robots and their influence on customers (Belanche et al. 2021; Mariani et al. 2023; van Doorn et al. 2017), literature can still benefit from further empirical research (Lu et al. 2020). Existing empirical studies (e.g. Holthöwer and van Doorn 2022; McLeay et al. 2021) are rarely based on responses to actual interactions with service robots (exceptions e.g. (Schepers et al., 2022; Steins et al., 2024)).

Scenario studies may not evoke sufficient emotional involvement needed for participants to fully engage with and respond to the situation (Kim and Jang 2014). This limitation is especially pertinent for service robots, as many customers have limited direct experience with them and may instead draw on inaccurate media portrayals. Thus, in addition to two online studies, I consider customers' first-hand interactions with a service robot through an observational laboratory study. Across all studies, I used a Furhat robot providing customer-facing services in two contexts: hospitality, where it recommended drinks, and retail, where it invited customers to try beverages in different cups and gathered customer feedback. This robot is embodied as a bust with limited physical mobility but is capable of advanced social interaction. Powered by GPT-3 and ChatGPT ("Integrating with OpenAI - Furhat Developer Docs" n.d.), it operates with high degree of autonomy. Its embodiment and autonomy qualify it as a service robot under the definition of Jörling et al. (2019).

This research contributes to the service literature in several ways. First, I demonstrate the importance of social presence for customer satisfaction and the mediating effect of service innovativeness. Second, I provide evidence that the effects of social presence vary depending on perceived robot characteristics and customer characteristics. Thus, this research contributes to understanding customer reactions to service robots (Hollebeek et al. 2019; Lu et al. 2020), and showcases the complementary role social presence and service innovativeness play in customer satisfaction. Third, I take an alternative theoretical perspective using cognitive dissonance (Festinger 1957) and schema theory (Bartlett 1995; Piaget 1952) to understand customers' reactions when interacting with service robots. Finally, my findings offer practical insights for designing services using service robots that enhance customer acceptance and satisfaction. Understanding customers' perceptions on service provision by service robots should enhance effective robot deployment and ultimately lead to a faster proliferation throughout the service industry.

2. Theoretical framework and hypotheses

2.1. Service robots in service encounters

Given the interest in physical service encounters and to differentiate them from virtual applications, I adopt the definition of service robots as highly autonomous, physically embodied information technology that provides service tasks (Jörling et al. 2019). Over time, service robots have evolved from executing simple routines to engaging in autonomous decision-making and frontline interactions (Borghi et al., 2023). As they increasingly replace or complement human employees, research has focused on factors influencing customer responses in robot-mediated service encounters. Table 1 presents examples of empirical studies examining customers' responses to service robots and the psychological processes through which characteristics of the robot or the robot–customer interaction shape those responses.

Prior studies highlight robots' impact on customer satisfaction, experience, behavioral intentions, perceived responsibility of negative outcomes, or attitude toward service (Borghi et al. 2023; Grazzini et al. 2023; Jörling et al. 2019; Mende et al. 2019). Beyond implementation of the robot, research (Grazzini et al. 2023; So et al. 2023; Yam et al. 2021) investigates the qualities of the robot and contextual factors that allow for a more nuanced understanding of the service and customer outcomes. Anthropomorphism and the human-like appearance of the service robot have received particular attention due to contradictory findings of their impact on customer satisfaction and the uncanny valley effect (Grazzini et al. 2023; Holthöwer and van Doorn 2022; So et al. 2023; Yam et al. 2021). Understanding the role of anthropomorphism in social engagement has been described as "the 'holy grail' of robotics" (El Halabi and Trendel 2024, p. 1). Relatedly, studies consider other "human" aspects of robots, such as social presence, which may be affected by robot anthropomorphism (Holthöwer and van Doorn 2022). Social presence of the robot was shown to influence customer perception of social judgment, engagement and loyalty (Čaić et al. 2019; Grewal et al. 2020; Holthöwer and van Doorn 2022; van Doorn, Mende, et al. 2017), yet it is often treated rather tangentially in empirical research (e.g., Choi et al. 2021; Grazzini et al. 2023; Jörling et al. 2019).

Given the social character of service encounters, prior research has drawn on social and cognitive psychology (e.g. theory of mind or social identity theory) to explain customer responses to service encounters with service robots (Holthöwer and van Doorn 2022; Mende et al. 2019; Yam et al. 2021). Similarly, I draw on psychology applying cognitive dissonance and schema theory (Festinger 1957; Fiske and Linville 1980), which has not yet been used in this context (Aggarwal and McGill 2007; Hall and Andriani 2003). Those theories allow for feasible explanation of the customers' perceptions and behaviors, which are dynamic enough to account for rapidly changing customer acquaintance with, and development of, service robots.

In this paper, I study the effect of social presence on satisfaction and service innovativeness while considering the impact of factors such as anthropomorphism and customer innovativeness in affecting those relationships. Fig. 1 presents a conceptual model tested further in the paper, which argues that customer responses to service encounters are fundamentally shaped by schema-consistency processes involving two co-activated cognitive structures: the service encounter schema, derived from repeated human-to-human service interactions, and a broader technology schema, which originates from technological artefacts present in people's life. When those established schemas are significantly violated, customers experience cognitive dissonance, which they resolve

Table 1
Overview of selected empirical studies of service robots in service encounters.

Source	Theory	Method	Key variables	Findings
Jörling, Böhm and Paluch (2019)	Attribution theory	Interviews; vignette experiments	Technology autonomy; perceived behavioral control; interaction option; valence of service; outcome responsibility	Autonomy of technology reduces perceived responsibility for positive but not negative outcome by decreasing perceived behavioral control. Perceived ownership increases perceived responsibility for negative outcome. Possibility to interact increases perceived behavioral control and perceived responsibility for positive outcome.
Mende et al. (2019)	Self-completion theory	Vignette experiments with photo and video stimuli	Compensatory behavior; robot vs. human	Humanoid robots lead to greater eeriness and identity threat, which leads to compensatory behavior. Social belonginess and healthfulness of food ease the discomfort triggered by the robot.
McLeay et al. (2020)	Brand congruency theory; socially responsible consumption	Vignette experiments	Role; robot type; perceived innovativeness; perceived ethical/societal responsibility fit; service type; value creation model; service experience; brand usage intention; openness to change; preference for ethical or responsible business	The role of substitution positively impacts service experience, and this is mediated positively through perceived innovativeness and negatively through perceived ethical/societal reputation.
Choi, Mattila and Bolton (2021)	Social cognition; theory of mind	Scenario-based experiments:picture-based animation-based	Robot anthropomorphism; warmth; satisfaction; failure; recovery	Humanoid robots are associated with greater warmth. Yet customers are more dissatisfied with failure of humanoid robots, but through the service recovery, such robots can improve satisfaction.
Yam et al. (2021)	Theory of mind	Field experimentlab experiment	Anthropomorphism; perceived agency; perceived experience; customer satisfaction	Anthropomorphism leads to satisfaction, but this relationship is mediated by perceived experience.
Chang and Kim (2022)	Theory of social perception	Vignette experiments: scenario-based and video based	Satisfaction with the robot; relationship orientation; robot type; warmth; competence	Satisfaction with the functional and social service robots differs for firm relationship orientation. This relationship is mediated through warmth for social service robots and competence for functional service robots.
Holthöwer and van Doorn (2022)	Social identity theory	Online, lab and field experiment	Robot anthropomorphism; automated social presence; social judgment; embarrassment; behavioral intention	Customers feel less judged by robots (vs. human) in embarrassing situations. However, robots high anthropomorphism is associated with higher automated social presence and social judgment.
Borghi et al. (2023)	sRAM; construal level theory	Online reviews and analysis	Customer satisfaction online rating; rapport with the robot; device deployed to post; customer review effort focused on the robot; online review experience	Rapport with robots increases the extent of the relationship between the robot and customer satisfaction. High effort focused on robot during writing the review decreases the extent of the relationship between robot and satisfaction.
Grazzini, Viglia, and Nunan (2023)	Expectation disconfirma-tion theory; social cognitive theory	Focus group; laboratory experiment	Robot anthropomorphism; eeriness; disconfirmation of expectations; warmth and competence; attitude toward the service	High human likeness causes more negative customer responses. It is mitigated if the robot shows warmth rather than competence.
El Halabi and Trendel (2024)	Psychological ownership	Online experiment, laboratory experiment	Perceived control; perceived familiarity; eeriness; interaction enjoyment; repurchase intention	By the act of naming the service robots, customers' perception of eeriness decreases, allowing for greater enjoyment from the interaction and subsequently leaves customers with intention to repurchase the service.
This study	Cognitive dissonance theory; schema theory	Online studies; laboratory observational study	Social presence; service innovativeness; customer satisfaction; customer innovativeness; robot anthropomorphism	Social presence positively affects customer satisfaction with the robot. This relationship is mediated by service innovativeness. For less innovative customers, service innovativeness does not affect satisfaction. For high robot anthropomorphism, social presence impacts satisfaction only through the service innovativeness.

through perceptual adjustments that are shaped by social presence and interpreted through innovativeness-related meaning making.

2.2. Social presence

Originally, social presence was not associated with nonhuman actors. When introduced, social presence was defined as the salience of interaction partners and their interpersonal relationships during mediated interaction (Short et al. 1976). The concept was heavily applied in computer-mediated interactions, even though originally, both interaction partners were human beings. The main component of social presence has been considered "intimacy", which is understood as the feeling of connectedness and immediacy, representing the psychological distance between parties (Short et al. 1976).

Van Doorn et al. (2017) introduced the notion of automated social presence to tailor the construct to nonhuman interaction partners. With the rise of artificial intelligence, people have started to engage in quasi-social relationships with it in different forms. Thus, the meaning of social presence has evolved to be applied also to, among others, human-to-robot interactions concerning the feeling of being in someone else's company (Heerink et al., 2008). Social presence in human–robot interactions pertains to the psychological state in which the robot is experienced or perceived as a social actor (van Doorn et al. 2017; Lee 2004). This enabled people to form relationships with the service robots, thereby enhancing relational experience from social interactions (Lee et al. 2006).

Until recently, it was not necessary to extend understanding of social presence to robots, as customers interacted solely with frontline employees in service encounters. Thus, customers developed a schema of service encounters in which they interacted with human beings. Schema refers to "cognitive structures of organized prior knowledge, abstracted from experience with specific instances; schemas guide the processing of

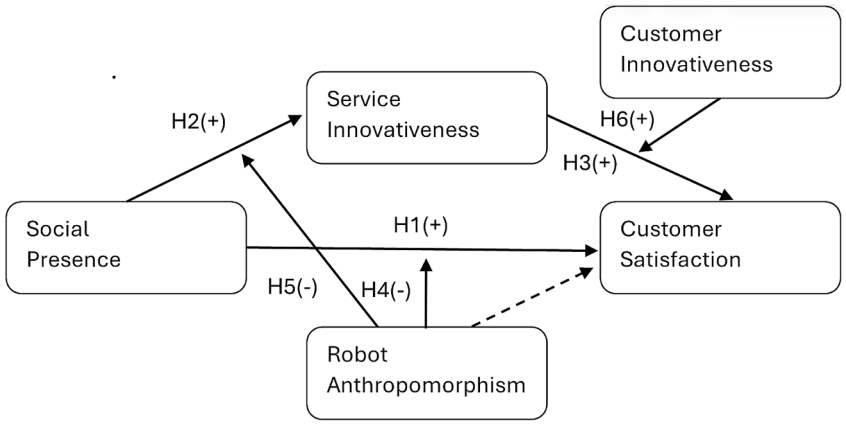


Fig. 1. Conceptual framework.

new information and the retrieval of stored information" (Fiske and Linville 1980, p. 543). Schema is activated consciously or unconsciously by an idea or concept, and so all other associated information is accessed (Fiske and Linville 1980). Service encounters that include interaction with a robot instead of a human employee are considered different from "ordinary". Service encounters follow well-established scripts that guide both employees and customers as to their roles and expectations toward the service (Mohr and Bitner 1991; Solomon et al. 1985). The script is a particular type of schema that focuses on the sequence of events (Schank and Abelson 1977). Service script is learned based on one's previous experiences with services and allows customers to navigate in service encounters with minimal effort (Järvi et al. 2020; Solomon et al. 1985). As they are based on repetitive past encounters, service scripts would often assume that the fellow actor following the script is human, as that is what most service experiences in the past have been. Entering service encounters, customers often apply the same schemas, or in this case, script, to interact with robots as they would in the case of human-to-human interaction (Söderlund 2021).

Yet, the presence of robot instead of human in service encounter causes mismatch in the service encounter schema and the actual situation and may lead to cognitive dissonance, which manifests in psychological discomfort (Festinger 1957). Discomfort is proportional to the magnitude of the dissonance (Festinger 1957). However, if the robot is considered a social entity, it can partially reduce this mismatch because the robot resembles the expected social interaction partner. With higher social presence of a robot, the service encounter bears greater resemblance to the service encounter schema. In turn, the magnitude of dissonance is lower and so is the discomfort. With less or no discomfort, consumers can more fully experience benefits of the service. Thus, I propose that:

H1: The level of social presence of a service robot is positively associated with customer satisfaction with a service encounter.

2.3. Service innovativeness

While social presence may as hypothesized reduce cognitive dissonance between experiences and received service, novel and unusual experiences, similarly to novel products, are not always easy to comprehend (Mugge and Dahl 2013). That is why it is crucial to consider customers' perceptions of implemented service innovations (Pilawa et al. 2022). Service innovativeness can be helpful in processing novel service experiences as it is in very essence concerned with such occurrences.

McLeay et al. (2021) considered service innovativeness in the context of service robots and defined it based on Hurley and Hult (1998) and Kunz et al. (2011) – thus, as a customer's perception of a service provider's propensity to launch novel and creative solutions. Such understanding implies that the providers are implementing novel services, or to put it differently, service innovations. However, those service innovations implemented by the provider need to be perceived as novel if the provider is to be viewed as innovative (Lee et al. 2009). In contrast, I conceptualize service innovativeness on product level as the customer's evaluation of the meaningfulness and novelty of service in comparison to alternatives (Mu et al. 2018; Zolfagharian and Paswan 2008). Here, service innovativeness is an attribute of the service offering, yet its degree varies between services (Zolfagharian and Paswan 2008). Thus, I follow this conceptualization, as I believe that the implementation of a service robot signals more about the service than the provider whose other services may be far from innovative.

The sole introduction of the robot may already be enough to consider service innovativeness, yet I wish to focus particularly on the impact of the social presence of the robot on service innovativeness. While social presence may ease the dissonance connected with the service encounter schema (H1), it may as well create a different dissonance in relation to other schemas. Since social presence is an inherently human-associated characteristic, its higher level in nonhuman objects may go against the pre-existing knowledge and schemas concerning technology and its distinction from humans. When cognitive dissonance becomes overbearing, people strive to reach comfort and minimize the mental burden (Aronson 1969; Festinger 1957).

One way to achieve cognitive comfort is by revising existing schemas. Thus, cognitive dissonance triggers the cognitive process of adaptation, which consists of assimilation or accommodation (Piaget 1952). Assimilation is a process of utilizing mental effort to fit new information into existing schema (Bingham and Kahl 2013). Accommodation, however, occurs when there is no schema similar enough to use for navigating one's understanding of new information, in this case, the existing schema is changed, or a new schema is created (Taylor and Uchida 2019). Piaget (1952) argues that both of those processes are interdependent and happen simultaneously to a different extent when people face unfamiliar situations. It is easier to assimilate or accommodate new information if there is some familiarity added through analogies and associations. In non-threatening contexts, those processes represent constructive meaning-making (Harmon-Jones et al. 2009). Technology and innovativeness are often associated with each other (Hall and Andriani 2003) making service innovativeness a plausible interpretive framework through which customers can accommodate the unexpected social presence of a robot. Service innovativeness allows customers to view the robot's unexpected social capabilities as evidence of a novel and meaningful service offering rather than merely as an inconsistency.

While the dissonance created by social presence is uncomfortable, social presence itself is associated with engagement, warmth, and interpersonal connection (Lee et al. 2006; van Doorn et al. 2017). This interpretation provides a cognitively economical way of restoring consistency.

This reasoning is consistent with the suggestions of Moreau et al. (2001) as well as Mugge and Dahl (2013) that customers evaluate new products by trying to incorporate them into their existing schema of similar products, and for a more innovative product, customers experience greater differences between the schema and the actual state. Thus, customers would reduce the dissonance by making the novel situation fit more closely into the existing or slightly adjusted schema connected with technology. In such a way, customers legitimize the unexpected social presence while maintaining coherent beliefs about technology. Consequently, service innovativeness helps in dissonance reduction, transforming initial discomfort into favorable evaluations.

Moreover, research provides evidence that service innovativeness is positively connected to satisfaction in usual service encounters (Stock 2011). Considering the presented reasoning, I propose that the relationship between social presence and satisfaction is mediated by service innovativeness. Thus, I propose that the higher is social presence the higher is also service innovativeness, and in turn, satisfaction with service encounter. Accordingly:

H2: The social presence of the service robot is positively associated with service innovativeness.

H3: Service innovativeness is positively associated with customer satisfaction with service encounters.

2.4. Anthropomorphism

Anthropomorphism refers to ascribing human characteristics to nonhuman entities (Epley et al. 2007). This concept has been used in marketing and service literature for a long time in various contexts, such as cars, brands, etc (Aggarwal and McGill 2007; Puzakova and Aggarwal 2018). However, it became especially prominent in literature when considering human–robot interactions (Blut et al. 2021; Kim et al. 2019). Robot anthropomorphism sometimes refers to the degree of its human-like design (Holthöwer and van Doorn 2022), while it is also defined as a degree to which "customers perceive service robots as humanlike, rather than to the extent to which firms design robots as humanlike" (Blut et al. 2021, p. 633). These perspectives are distinct yet strongly connected, as human appearance facilitates customers to perceive robots as human-like (Broadbent 2017). In this paper, I follow the latter conceptualization of anthropomorphism as customer perception of service robots as humanlike, given my interest in customer perception. Despite both anthropomorphism and social presence being consumer perceptions, they are conceptually distinct. While anthropomorphism reflects perception of the robot itself based on cues such as appearance or form, social presence reflects the experience of interaction with the robot.

The importance of anthropomorphism shaping customer evaluations of service robots is widely recognized (see Table 1). However, it is also controversial, with studies presenting contradicting arguments for the positive influence of robot anthropomorphism on the customer, such as satisfaction (So et al. 2023) or intention of use (Blut et al. 2021), as well as negative results, such as lower levels of liking (Kim et al. 2019), lower satisfaction (Crolic et al. 2022), as well as compensatory behavior (Mende et al. 2019) proving the hypothesis of the uncanny valley effect. Prior research generally finds a positive association between anthropomorphism and satisfaction (Blut et al., 2021; Söderlund & Oikarinen, 2021; Qiao & Song, 2024; van Doorn et al., 2017). I include this established relationship to account for prior findings. However, explaining the direct effect of anthropomorphism is not the objective of this research. Accordingly, anthropomorphism is theorized instead as a contextual condition that shapes the extent to which social presence influences both satisfaction and service innovativeness.

Anthropomorphism and social presence also differ in terms of when and how they are formed. Anthropomorphism can be inferred rapidly and even prior to interaction based on visual and design cues (Bartneck et al. 2009; Blut et al. 2021). Social presence, however, relates to more subtle cues like psychological distance or nonverbal cues (Walther 1992), which are not easily observable and emerges only during the interaction as individuals respond to the robot's communicative and relational behavior. Accordingly, anthropomorphism operates as a pre-interaction cue, whereas social presence is experience-based and develops throughout the interaction.

Following a particular schema based on previous experiences, a customer navigates the service encounter (Järvi et al. 2020; Solomon et al. 1985), and may expect human as interaction partner. The mismatch between schema and actual situation leads to cognitive dissonance, the discomfort which one aims to minimize (Festinger 1957). Thus, anthropomorphism can reduce initial schema violation before social interaction unfolds. That may weaken the urgency of reducing the dissonance when social presence is subsequently experienced. This implies that higher anthropomorphism decreases the influence of social presence on customer satisfaction, meaning that anthropomorphism acts as quasi-moderator in the relationship between social presence and consumer satisfaction.

H4: Robot anthropomorphism negatively moderates the positive relation between social presence and customer satisfaction with a service encounter.

Robot anthropomorphism may also interact with expectations regarding technology and human boundaries. People want to distinguish between human and nonhuman entities (Wiese and Weis 2020). Thus, the higher level of robot anthropomorphism may challenge pre-existing schemas concerning technology and its distinction from humans, causing cognitive dissonance, consistent with the uncanny valley effect (Mori 1970). This dissonance, which is considered to be especially uncomfortable (Blut et al. 2021; Mende et al. 2019), may become a primary cognitive focus , reducing the relative importance of social cues emerging later during interaction. Consequently, the dissonance arising from the robot's social presence may become secondary. Thus, the impact of social presence on service innovativeness may be weaker for greater robot anthropomorphism, as cognitive resources are allocated to resolving schema inconsistencies at the level of anthropomorphism instead.

H5: Anthropomorphism negatively moderates the positive relation between social presence and service innovativeness.

2.5. Customer innovativeness

Higher service innovativeness can lead to greater customer engagement, patronage intention, and innovation success, yet this positive relation is often moderated by traits of individual customers (Avlonitis et al. 2001; Leckie et al. 2017; Zolfagharian and Paswan 2009). I would like to consider one of these traits - customer innovativeness, which is well-established theoretical construct in marketing and innovation literature (Cowart et al. 2008; Hirschman 1980; Midgley and Dowling 1978).

Customer innovativeness is a customer trait describing the "receptivity to new experiences and novel stimuli" (Im et al. 2003) and innate proclivity to adopt innovations (Midgley and Dowling 1978; Hirschman, 1980; Roehrich 2004). Such conceptualization is rooted in the psychological characteristics of individuals (Bartels and Reinders 2011; Steenkamp et al. 1999). Highly innovative individuals express their desire for novelty by accepting new ideas or adopting new products and services (Hirschman 1980). Such customers are rather insensitive to risk and more willing to seek new solutions (Steenkamp and Baumgartner 1992). Thus, it is understandable that innovative customers prefer innovative products – in fact, the fit between customer innovativeness and product innovativeness was found to positively influence satisfaction and behavioral intention (Cowart et al. 2008). I believe that in the

case of services, the fit between customer and service innovativeness should still positively influence consumer outcomes, especially since service innovativeness, particularly at high level, is supposed to more positively influence customer satisfaction than in the case of goods (Stock 2011).

From the perspective of cognitive dissonance theory, people experience a specific degree of cognitive dissonance differently. As argued by Festinger (1957), tolerance to cognitive dissonance depends on individual characteristics. While some individuals are unaffected by uncertainty and inconsistencies, others are highly sensitive to it, and this can be explained by individual differences and distinctive personality traits (Aronson 1969; Festinger 1957). Such factors include self-esteem (Aronson 1969), extroversion (Matz et al. 2008), or preference for consistency (Cialdini et al. 1995). People with a low preference for consistency are those who are tolerant of cognitive dissonance (Cialdini et al. 1995; Nail et al. 2001). Cialdini et al. (1995) describe people with a low preference for consistency as individuals who are receptive to novelty and open to stimuli that are new and so inconsistent with, for instance, previous beliefs or actions. Thus, the low preference for consistency is likely to correlate with (high) customer innovativeness, as both traits describe an individual's receptivity to novelty. Hence, more innovative customers will likely be more tolerant toward cognitive dissonance as they tend to prefer the new and unknown, which shows that they are not bothered with inconsistency; rather it is the opposite – they enjoy it.

If innovative customers are less sensitive to cognitive dissonance than less innovative customers, then greater inconsistencies are needed for them to feel the psychological discomfort and to return to the state of comfort if discomfort occurs. Thus, I propose that when customer innovativeness is high, the relationship between service innovativeness and satisfaction is strengthened.

H6: Customer innovativeness positively moderates the positive relation between service innovativeness and customer satisfaction with the service encounter with service robots.

3. Overview of studies

I conducted two online and one laboratory study to test the conceptual framework detailed in Fig. 1. All studies employ the same service robot with the same mask (face appearance) and voice to control for potential effects on social presence and other variables. I used a Furhat robot that is in the physical form of a sculptor's bust. Despite lacking locomotion, its head mobility, embodiment, and autonomy qualify it as a service robot under the definition of Jörling et al. (2019). Appendix A (Fig. A1) contains the photo of the laboratory study set up presenting the used robot. Table 2 presents a summary of all three studies. Across all studies, robot provided customer-facing services in two contexts: hospitality (Study 1 and 2) and retail (Study 3).

Study 1 tests my thesis linking robot social presence to customer satisfaction and service innovativeness, which in turn impacts customer satisfaction. Study 2 replicates the original results and further tests the moderating effect of robot anthropomorphism on social presence and satisfaction, as well as on the effect between social presence and service innovativeness.

Studies 1 and 2 were online surveys conducted using Connect, a platform that facilitates access to panel respondents, and the sample consisted of U.S. customers. In contrast, Study 3 involved real-life interaction between participants and robot exemplifying a service encounter. Unlike in the previous studies, respondents were Swedish, which enhances the generalizability of the findings. I use the study to re-analyze the relations of the core model (H1, H2, H3) and test the moderating role of customer innovativeness on the relation between service innovativeness and customer satisfaction. As customer innovativeness is self-assessed, it is more accurately measured in real-life settings when facing situations in which given features are important.

Although the order of the measurement scales differed across studies, it was not randomized within each study. I include a correlation matrix with square roots of average variance extracted (AVE) in Appendix B, demonstrating good discriminant validity based on the Fornell-Larcker criterion (Fornell and Larcker 1981). Additionally, the HTMT ratio values are below 0.90. Hence, I can conclude acceptable discriminant validity.

Finally, in Appendix A (Table A1 and A2) I present composite reliability, and Cronbach alpha values, all of which are above 0.70 and mostly above 0.80, indicating good reliability. Convergent validity is also on a good level since the AVE values are greater than 0.7 for all constructs across all studies.

Table 2
Overview of studies.

Study	Main constructs	Tested hypothesis	Type of study	Sample size
Study 1	Social presence, service innovativeness, satisfaction	H1, H2, H3	Online	245
Study 2	+ Anthropomorphism	H1, H2, H3, H4, H5	Online	300
Study 3	+ Customer innovativeness	H1, H2, H3, H6	Laboratory	77

3.1. Study 1 – social presence, service innovativeness and satisfaction

This study tests H1, H2, and H3.

3.1.1. Methods

I gathered 250 responses, out of which 245 passed the attention check and were used for the analysis. As attention check I implemented a question asking them to choose "strongly agree" from the provided options in both online studies. The sample consists of 51% female and 49% male. Most of the respondents were in their 20s (22%), 30s (30%), and 40s (26%). Additionally, around 46% of the respondents had a bachelor's degree and 16% were college graduates.

Participants were asked to watch a short (35 sec) video of the service robot performing frontline hospitality service welcoming them as customers in a café, recommending seasonal coffee, and asking them to place an order. I include the study video as supplementary material. The same video was used for Study 1 and Study 2 to eliminate potential influence of changes in the verbal, visual, and audial cues. After watching the video, participants were asked to answer the survey.

I used multi-item scales to measure the four central constructs. The items of each construct for all three studies are listed in Appendix A (Table A1). Satisfaction was measured using three items from Söderlund (2016) on a scale from 1 to 10. All remaining constructs were measured on a Likert-type scale from 1 to 7, where 1 indicates "highly disagree" and 7 "highly agree". I have not adjusted satisfaction scale to 1-7 scale since for OLS regression, it is not necessary as only the values of coefficients are affected by it but not the test statistics (Wooldridge, 2006). To capture social presence, I used a six-item measure used by Gefen and Straub (1997, 2004). Service innovativeness was measured using four items inspired by the scales on product innovativeness used by Lowe and Alpert (2015) and Kim et al. (2015). Table 3 presents the descriptive statistics for variables used in studies.

3.1.2. Results

To test the hypotheses H1, H2, and H3 I ran a mediation analysis using PROCESS macro (model 4) for SPSS (Hayes 2022). Social presence served as an independent variable, customer satisfaction was a dependent variable, and service innovativeness was a mediator. I included gender, age, and income as control variables; however, the results remain substantively unchanged with their inclusion. The results show a significant positive direct effect of social presence on customer satisfaction ($\beta = 0.96$, $p < 0.001$). Thus, H1 is supported, which means that

for higher social presence of the service robot, customer satisfaction with the service encounter is also higher. The analysis also confirmed (H2) that social presence has a significant positive effect on service innovativeness ($\beta = 0.45$, $p < 0.001$). That indicates that the higher the social presence of the robot the higher the customer perception of service innovativeness. Finally, results show that service innovativeness had a positive effect on customer satisfaction ($\beta = 0.50$, $p < 0.001$). Thus, H3 is supported, meaning that customer satisfaction is associated with service innovativeness. The total indirect effect was found to be positive and significant (Effect = 0.23; CI [0.15, 0.31]). The significance of the indirect effect was evaluated using bootstrapping with 5000 samples. The results suggest that there is a partial mediation of the effect of social presence on customer satisfaction with service encounters through service innovativeness.

3.2. Study 2 – Anthropomorphism

This study tests H1, H2, H3, H4 and H5

3.2.1. Methods

To further replicate the results, enhance the credibility and validity of my findings, and test H4 and H5, I conducted an additional online study using an identical design to Study 1. In addition to already discussed measures, I included a measurement of anthropomorphism developed by Bartneck et al. (2009), which consists of five items measured on a semantic differential scale. As anthropomorphism and social presence both concern similar perceptions of the robot, the overlap is visible in the used scales as well as conceptual aspects. Yet they perform very different roles in this study.

The sample consisted of 300 responses, out of which 49.3% were women and 50.3% were men. Most respondents were in their 20s (32%), 30s (35%), and 40s (20%). Most of the respondents had a bachelor's degree (49%), graduated from college (15%), or held a professional degree (13.7%).

3.2.2. Results

As in the previous study, I used regression-based analysis (PROCESS macro, model 8; Hayes 2022) to replicate the main effects (H1, H2, H3) and investigate the quasi-moderation and moderation of anthropomorphism hypothesized in H4 and H5. I included gender and age as control variables; however, the results remain substantively unchanged with their inclusion. Firstly, I replicated the results from Study 1 (H1) showing the positive effect of social presence on customer satisfaction with a service encounter ($\beta = 1.36$, $p < 0.001$) as well as the mediation of this relation by service innovativeness as predicted in H2 ($\beta = 0.43$, $p = 0.01$) and H3 ($\beta = 0.47$, $p < 0.001$).

Considering the hypothesized moderation of the relation between social presence and service innovativeness by anthropomorphism, results showed that the direct effect of anthropomorphism on service innovativeness is not significant ($\beta = 0.32$, $p = 0.08$), nor is the interaction effect of anthropomorphism on social presence ($\beta = -0.08$, $p = 0.08$). Thus, H5 is rejected as the anthropomorphism does not moderate the relation between social presence and service innovativeness. However, the direct effect of anthropomorphism on satisfaction was significant ($\beta = 1.88$, $p < 0.001$), replicating results of prior research (Blut et al. 2021; Söderlund and Oikarinen 2021; Qiao and Song 2024; van Doorn et al. 2017). The anthropomorphism was also found to be a quasi-moderator for effect of social presence on satisfaction as hypothesized by H4, as the interaction effect was significant and negative ($\beta = -0.35$, $p < 0.001$). This suggests that with a higher level of anthropomorphism of the robot, the social presence has less positive effect on customer satisfaction with service encounters. To further understand the nature of anthropomorphism influence on the effect between social presence and satisfaction, I used a floodlight analysis (Arsenovic et al. 2022; Spiller et al. 2013). I identified the Johnson-Neyman point (mean-centered value 1.11 equal to 3.45 mean value) which indicated that when anthropomorphism was below this value the effect of social presence on satisfaction is significant, yet for higher values (indicating very high anthropomorphism) the effect is not significant. Fig. 2 shows that there is one more point (mean-centered 2.79) after which the effect again becomes significant, yet as this value is above the highest value on the used scale, I do not address it. As presented in Fig. 2, the results indicate that the rise of anthropomorphism (until the mean-centered value of 1.11) is associated with the decline of the positive effect of social presence on customer satisfaction with service encounters. Thus, for robots seen as more anthropomorphic the social presence has less of an impact on satisfaction. However, when the robot's anthropomorphism is very high (1.11 and above), the social presence no longer affects customer satisfaction with service encounters.

In line with those findings, the conditional direct effects between the social presence and satisfaction are significant for robots of lower anthropomorphism (for anthropomorphism = 1; effect = 1.01; CI [0.706 1.26]) but not for the robots of very high anthropomorphism (for anthropomorphism = 3.6; Effect = 0.10; CI [-0.13, 0.35]). That indicates that for robots with very high anthropomorphism, there is a full mediation, meaning the social presence influences customer satisfaction with service encounters only through service innovativeness.

3.3. Study 3 – Customer Innovativeness

This study tests H1, H2, H3 and H6.

3.3.1. Methods

To further replicate the main effects (H1, H2, H3) in a more realistic setting and test H6, I ran an observational laboratory study. The study was conducted at a shopping facility of one of the biggest Swedish retail chains. Participants were randomly recruited from customers entering the store. They were asked to sit in front of a service robot and interact with it for a few minutes to create a situation resembling a service encounter. The service robot provided an amenity service inviting participants to try refreshments in two different cups (paper and plastic) and engaging them in a short conversation about the beverage of choice, asking for their opinions about the beverage's taste. The choice of the cup was not a form of manipulation, and it should not impact satisfaction as each consumer was choosing according to their liking and in line with their preferences. The setup of the study is presented in Appendix A (Fig. A1). Participants were left alone with the robot for the time of the service encounter, after which they were asked to answer a survey.

I gathered 77 full responses. The sample consisted of a nearly equal number of males (49.4%) and females (50.6%). Most of the respondents

Table 3
Descriptives.

Study		Social presence	Service innovativeness	Satisfaction	Anthropo-morphism	Customer innovativeness
Study 1	Mean	2.77	4.94	5.27		
	SD	1.46	1.43	2.38		
Study 2	Mean	3.10	5.28	5.92	2.31	
	SD	1.64	1.25	2.50	1.07	
Study 3	Mean	3.20	5.21	6.80		5.33
	SD	1.47	1.44	1.93		1.25

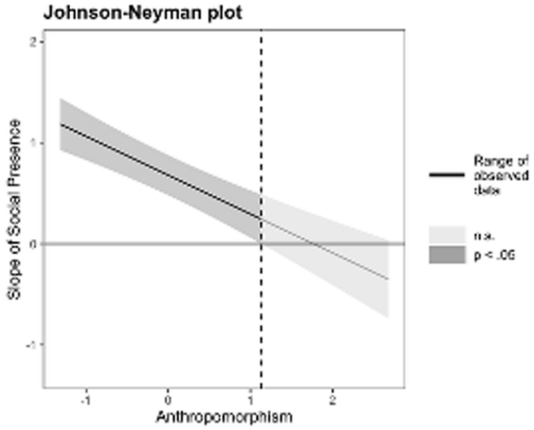


Fig. 2. Johnson-Neyman plot for anthropomorphism moderation.

were in their 20s (24.7%) and 30s (23.4%). Additionally, around 32.5% of respondents were secondary school graduates, 31.2% had attended university without graduation, and 26% had obtained a bachelor's degree.

Given the in-person setting and expected smaller sample, anthropomorphism was not measured in this study to keep the questionnaire concise. Instead, a measure of customer innovativeness was added to the original set of measures. It uses the Likert scale (1–7) and consists of four items based on selected items proposed by Pallister and Foxall (1998), who shortened the scale of Hurt et al. (1977). Since the items are negatively phrased they were reverse-coded before the analysis to facilitate interpretation of the results.

3.3.2. Results

To test the hypotheses (H1, H2, H3, H6) I ran a moderated mediation using PROCESS macro (model 14) (Hayes 2022). I used the basic model from Study 1 while adding consumer innovativeness as moderator of the link between service innovativeness and satisfaction. I included gender, age, and previous interaction with the robot as control variables; however, the results remain substantively unchanged with their inclusion. The analysis supported H1 regarding the positive association between social presence and satisfaction ($\beta = 0.53$, $p < 0.001$). H2 predicting positive link between that social presence and service innovativeness was also supported ($\beta = 0.42$, $p < 0.001$). Finally, the connection between service innovativeness and customer satisfaction was not significant ($\beta = -0.88$, $p = 0.18$) in the presence of the moderator (H3 was rejected). However, that does not indicate that the results of previous studies are not replicated, rather, it shows the importance of the moderator.[1]

Despite the lack of a significant effect of service innovativeness on satisfaction, the interaction effect of service innovativeness and customer innovativeness was significant ($\beta = 0.24$, $p = 0.04$) supporting H6. This means that the link between service innovativeness and satisfaction is moderated by customer innovativeness. Consumer innovativeness does not have a significant direct effect on satisfaction ($\beta = -1.32$, $p = 0.06$), thus it is a pure moderator. As in Study 2 I used a floodlight analysis to further understand the existing moderation (Spiller et al. 2013). I identified the Johnson-Neyman point (mean-centered value -1 equal to 4.33 mean value), which indicated for what level of customer innovativeness the coefficient of the moderated relationship between service innovativeness and customer satisfaction is significant. As presented in Fig. 3, the results indicate that for more innovative customers (customer innovativeness above the mean-centered value of -1.00), service innovativeness is positively linked to customer satisfaction, while for customers who are less innovative (customer innovativeness below mean-centered value of -1.00), the service innovativeness is not linked to customer satisfaction. More innovative customers consider service innovativeness when assessing satisfaction with a service encounter, while non-innovative customers do not. The individuals with higher customer innovativeness (above the calculated threshold) constitute 81% of the sample.

In line with the moderation analysis, the conditional indirect effects between the social presence and satisfaction are significant for more innovative customers (for customer innovativeness = 5.25; effect = 0.16; CI [0.02, 0.33]; Customer innovativeness = 6.75; Effect = 0. 31; CI [0.06, 0.53]) but not for less innovative ones (for customer innovativeness = 3.87; effect = 0.02; CI [-0.18, 0.29]). Thus, the effect of social presence is partially mediated through innovativeness only for more innovative customers.

4. General discussion

Across three studies (see Table 4), I demonstrated that social presence plays a greater role in service encounters with service robots than was previously accounted for in the literature. Social presence is positively connected with customer satisfaction, especially for robots perceived as less anthropomorphic. Moreover, social presence is also positively connected to service innovativeness – thus it may help to shape the perception of the provided services. In turn, service innovativeness is positively related to customer satisfaction, especially for more innovative customers. That would mean that for more innovative customers, social presence has an even greater effect on customer satisfaction as it additionally influences it through service innovativeness. This mediation effect is of special importance when one considers highly anthropomorphic robots as, in such cases, social presence has no direct effect on satisfaction.

4.1. Theoretical implications

This paper responds to a call for research on robot service provision and its impact on customers (Hollebeek et al. 2019; Lu et al. 2020), by examining antecedents of satisfaction in service encounters with robots. My findings highlight the importance of the social presence of service robots in organizational frontlines as a determinant of customer satisfaction with service encounters. This strengthens previous claims by van Doorn et al. (2017), who propose to treat social presence as an important aspect of service robots, and further highlight the relevance of the social presence for customer outcomes.

The next contribution comes from exploring the relationship

[1] Results from running model 4 indicate that H3 was supported ($\beta = 0.41$, $p < 0.001$) and the previous results are replicated once the moderation is not considered in the analysis.

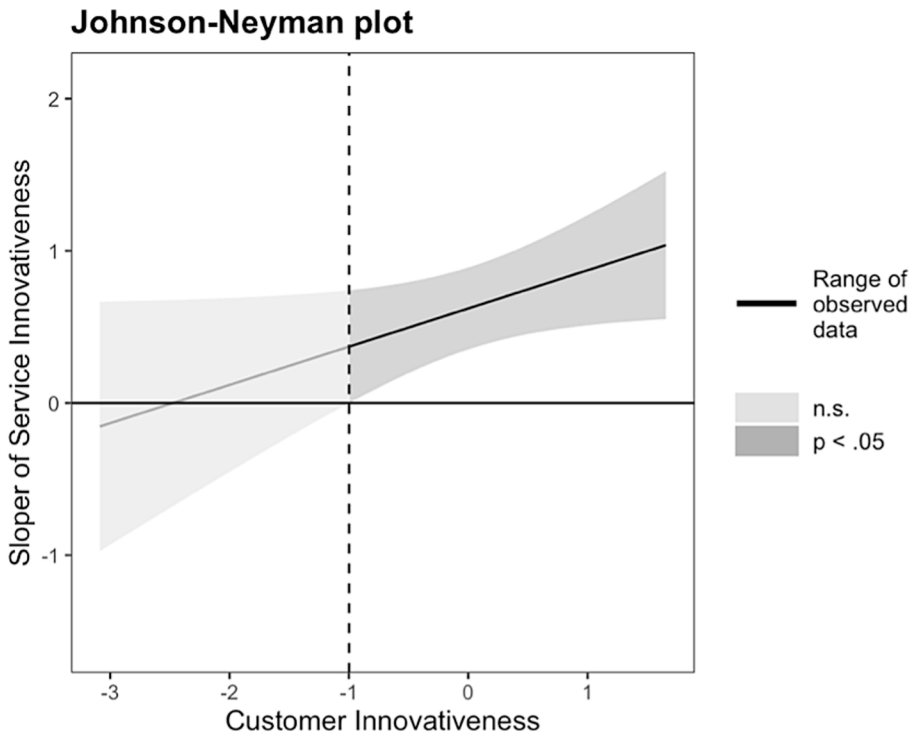


Fig. 3. Johnson-Neyman plot for customer innovativeness moderation.

between social presence and anthropomorphism. While findings reflect their conceptual proximity, as both relate to social cues, empirics further show they are distinct constructs. Unlike prior studies that primarily examine anthropomorphism and social presence as antecedents and consequences of one another (Blut et al. 2021; Holthöwer and van Doorn 2022; Noble and Mende 2023), my findings suggest a more nuanced relationship. Anthropomorphism functions as a boundary condition, showing that the positive effect of social presence on satisfaction diminishes for robots perceived as highly anthropomorphic. Most prior research focuses mainly on anthropomorphism while considering human-resembling features in robots, sparking the debate around inconsistent findings around the eeriness of human-resembling robots (Belanche et al. 2021; Blut et al. 2021).

However, it should not be the only aspect of interest. My findings emphasize the importance of considering anthropomorphism together with less observable aspects resembling humans such as social presence. Studies show that the feeling of eeriness is connected with inconsistencies in robot's realism or human resemblance (MacDorman and Chattopadhyay 2016). Consistency between design, realism, and social cues may help maximize benefits while reducing discomfort (El Halabi and Trendel 2024). Moreover, this is particularly important given that I found that social presence has a great influence on satisfaction as well as on service innovativeness.

This research also adds to the limited literature on service innovativeness. When considering service innovativeness, service research focuses on the provider's innovativeness rather than the innovativeness of a particular service that the provider offers (Dotzel et al. 2013; Kunz et al. 2011;Lee et al. 2009). Yet those are codependent, as the provider needs to introduce innovative services to be seen as innovative (Dotzel et al. 2013; Kunz et al. 2011). My findings indicate the importance of service innovativeness as the perceived degree of service novelty and meaningfulness by establishing that service innovativeness can enhance customer satisfaction (similarly to Stock 2011). For more innovative customers, there is a suggested positive effect of service innovativeness on satisfaction, yet in the case of less innovative customers, the effect seems to be absent, potentially due to sensitivity to uncertainty (Hirschman 1980). For them, the assimilation and accommodation of social presence into the technology schema through association with service innovativeness might not be strong enough of a dissonance reduction to allow them to reach the state of comfort.

Finally, while both schema theory and cognitive dissonance theory are not unknown in marketing (e.g., Aggarwal and McGill 2007), they have not been used to explain customer reactions to service robots. In contrast to dominant cognitive psychology approaches (Choi et al. 2021; Grazzini et al. 2023; Holthöwer and van Doorn 2022), my alternative theoretical perspective can help to better explain customers' reactions, as it accounts for the dynamic and malleable nature of human perception and the ever-changing environment. This is possible because these theories capture ongoing learning and the possible activation of multiple schemas.

Table 4
Summary of results.

Hypothesis	Relationship	Study 1	Study 2	Study 3
H1(+)	Social Presence -> Customer satisfaction	Supported	Supported	Supported
H2(+)	Social Presence -> Service Innovativeness	Supported	Supported	Supported [a]
H3(+)	Service Innovativeness-> Customer Satisfaction	Supported	Supported	Supported
H4(-)	Social Presence x Anthropomorphism -> Customer Satisfaction		Supported	
H5(-)	Social Presence x Anthropomorphism -> Service Innovativeness		Not Supported	
H6(+)	Service Innovativeness x Customer Innovativeness -> Customer Satisfaction			Supported

[a] When the analysis was run without moderation.

4.2. Managerial implications

Service providers no longer face the question of whether to adopt

robots but rather which type of robots to implement. My findings offer practical guidance for introducing service robots effectively. Specifically, service robots perceived to a greater extent as social entities lead to higher customer satisfaction. However, the benefits of social presence depend on how anthropomorphic they are perceived. Thus, one should also consider how customers may assess the anthropomorphism of the robot, as for robots seen as more anthropomorphic, the benefits of social presence are smaller. Thus, it would be wise to focus on social presence, particularly for robots that do not encourage customers to perceive them as highly anthropomorphic- often with more object-like appearance. Providers should ensure that robots are clearly perceived as social agents without strongly triggering anthropomorphism. This can be achieved through features such as conversational cues (e.g., vocal fillers; Goble and Edwards 2018) or more personalized communication (Gefen and Straub 2004).

Additionally, providers seeking to differentiate their offering by service innovativeness can benefit from deploying robots with high social presence, regardless of their anthropomorphism. Such robots enhance perceptions of service innovativeness, particularly among highly innovative customers, for whom this also increases satisfaction. When serving a more diverse consumer base, providers should support less innovative customers through education and gradual exposure (Bell et al. 2017). During early implementation, managers may also offer parallel human service options, directing robot services mainly toward more innovative customers.

Finally, managers should remain cautious, as high social presence may cause discomfort in sensitive or embarrassing service situations by increasing perceived social judgment (Holthöwer and van Doorn 2022).

4.3. Limitations and future research

As service robots become more widespread, existing schemas of service encounters may change or new ones may emerge, although schemas are generally resistant to change (Fiske and Linville 1980). Consequently, my explanations and findings may lose validity over time, highlighting the importance of studying how schemas evolve.

Moreover, I did not measure the magnitude of the dissonance or the change of the magnitude when reduction occurs, which could provide deeper insight, particularly among less innovative customers, for whom schema adaptation may be insufficient to substantially reduce cognitive dissonance.

Anthropomorphism was measured as a perceived attribute rather than manipulated, limiting its strength as a boundary condition. Moreover, both anthropomorphism and social presence were measured, thus, I cannot establish their temporal order. While the results support the proposed interaction between both constructs, future research should examine whether anthropomorphism also functions as an antecedent of social presence. Including realism checks could further strengthen the methodological rigor. The general innovativeness scale may also be overly broad, suggesting value in applying domain-specific measures. The scales measuring service innovativeness fail to capture the meaningfulness of the innovation which is an important element of the concept.

Future research should examine different industries to enhance generalizability, especially since robots are being implemented across various industries such as hospitality (Borghi et al. 2023), healthcare (Čaić et al. 2020), and financial services (Borghi et al. 2023). Given my focus on physical robots, further studies should also explore whether these findings extend to virtual agents such as chatbots. Moreover, the studied robot may differ from consumers' prototypical image of a robot, as its embodiment does not allow it to freely change location.

Finally, although Study 3 increases ecological validity by examining actual interactions with a service robot, its relatively small sample size limits the precision of estimated effects, thus, the findings would benefit from replication in larger field settings.

CRediT authorship contribution statement

Joanna Pilawa: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbusres.2026.116607.

Data availability

Data will be made available on request.

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